14.01 — Principles of Microeconomics

100 points

Problem Set 4 — Graded

Fall 2023

Student: Amin Cheipournejad · Total: 91.5 / 100

Grade Report

Part	Score	Grader's comment
Q1 1.1	**4.5 / 5**	False, with the right concept: a price control creates deadweight loss. Half a point because the justification overstates it and leaves out the mechanism. Only a binding control does any damage — a ceiling above the equilibrium price or a floor below it changes nothing — and the loss comes from trades that would have benefited both sides no longer taking place.
Q1 1.2	**4.5 / 5**	False, and the horizontal sum, P/3 + P/2 = 10P/12, is the right calculation: aggregate quantity is 5P/6, so aggregate demand is P = 1.2Q, not 5Q. Half a point because the working never says so, and the line next to it adding the two price equations, 2P = 5q, is the vertical sum that produces the statement's error, left on the page without comment.
Q1 1.3	**5 / 5**	True, matching the official answer: with free entry and identical firms, price equals marginal cost equals average cost, so economic profit is zero.
Q1 1.4	**1 / 5**	True is wrong. Free entry comes with free exit, and it drives the price to the minimum of average cost from either side: down when firms are making profits, up when they are making losses and some leave. The answer's argument, that blocking entry leads to market power and higher prices, covers only the first case. One point for that half.
Question 1 subtotal	**15 / 20**	

Part	Score	Grader's comment
Q2 2.1	**5 / 5**	Both elasticities correct, as general expressions, $-p/(4-p)$ and $-p/(2-p)$, and at the equilibrium prices, $-1/2$ and -1.
Q2 2.2	**5 / 5**	Correct, with the right reasoning: walkable Boston's residents have alternatives, so its demand is the more elastic second market, while car-dependent San Antonio has the less elastic first one.
Q2 2.3	**5 / 5**	Both equilibria correct: price and quantity of 4/3 in the first market and 1 in the second, worked at the top of the question and restated under 2.4.
Q2 2.4	**5 / 5**	Correct, though written out in the working for 2.6 rather than here: consumer and producer surplus of 16/9 and 8/9 in the first market and 1/2 each in the second, each computed and labelled as the surplus at the original equilibrium price.
Q2 2.5	**3 / 5**	The new prices and quantities are correct, 5/4 and 1/2 at the floor of 3/2, and so is the conclusion that the second market's quantity falls by more. The link to elasticity is backwards: the annotations put the first market's small change down to elastic demand and the second market's drastic change down to inelastic demand, when the second market is the elastic one, as the answers to 2.2 and 2.6 both say. The intuition goes no further than pointing back to those labels: with elastic demand, buyers have alternatives and cut back sharply when the price rises, so the floor removes more sales. Two points.

Part	Score	Grader's comment
Q2 2.6	**5 / 5**	Correct throughout, with every surplus computed by integration: in the first market consumer surplus falls by 31/144, producer surplus rises by 59/288 and the deadweight loss is 1/96; in the second, −3/8, +1/8 and 1/4. The interpretation is right as well: the second market's more elastic demand means the floor chokes off more trades, so it loses more.
Question 2 subtotal	**28 / 30**	

Part	Score	Grader's comment
Q3 3.1	**5 / 5**	Correct: each firm supplies q = p, and the market 5p. The working calls the aggregate “n times marginal cost”; what adds up is the five firms' quantities at each price, though the result is right.
Q3 3.2	**5 / 5**	Price 10 and quantity 50, consumer and producer surplus of 250 each, and 50 per firm, all correct, with the surpluses integrated.
Q3 3.3	**5 / 5**	Correct, matching the official answer: a firm would pay up to its whole profit of 50, which unrestricted entry would compete away. Against the limited entry of part 4, which leaves each firm 200/9, the most worth paying would be the difference, about 27.8 — a refinement the question does not require.
Q3 3.4	**4 / 5**	The new price, 20/3, and quantity, 200/3, are correct, as are the changes in consumer surplus, +1750/9, and total producer surplus, −250/9. The change for each individual firm, which the question also asks for, is not computed: each of the ten firms now earns 200/9, down from 50, a change that happens to equal the total's −250/9. One point.
Q3 3.5	**5 / 5**	False, and justified more fully than the official answer: entry moves surplus from producers to consumers, the reverse of what the statement says, while total welfare does rise, as the answer says — from 500 to about 667 in this example.
Question 3 subtotal	**24 / 25**	

Part	Score	Grader's comment
Q4 4.1	**2 / 2**	Price and quantity of 5000, consumer surplus of 12.5 million and producer surplus of 25 million. Correct.
Q4 4.2	**5 / 5**	Price 7000 at the same quantity, consumer surplus unchanged at 12.5 million and producer surplus of 35 million. Correct.
Q4 4.3	**5 / 5**	Correct, with the economic reason the question demands: with supply fixed, extra demand cannot raise the quantity, so it only raises the rent, and the whole subsidy ends up with landlords.
Q4 4.4	**4.5 / 5**	Price 3000, quantity 7000, consumer surplus of 24.5 million and producer surplus of 21 million, all correct. Half a point for the first line, which writes demand as 1000 − p; the next line uses 10000, so this reads as a slip.
Q4 4.5	**5 / 5**	Correct, with a good intuition on both sides: consumers gain through new renters entering and lower rents for existing ones, while homeowners lose because the extra supply lowers the rent they can charge.
Q4 4.6	**3 / 3**	Increasing supply, for the right reason: it lowers rents and houses more people, where the subsidy only raised the price.
Question 4 subtotal	**24.5 / 25**	

Report Card

Section	Earned	Available	Percent
Question 1 — True or False	**15**	**20**	**75%**
Question 2 — Minimum Price and Demand Elasticity	**28**	**30**	**93%**
Question 3 — Market Surplus with Firm Entry	**24**	**25**	**96%**
Question 4 — Market Surplus with Inelastic Supply	**24.5**	**25**	**98%**
Total	**91.5**	**100**	**92%**

Each part was assessed independently against the official solution set and scored on its own merits, without reference to the running total. Full marks were awarded wherever the work is correct and complete, and where the student's method departs from the official solution but is valid it earns full credit. Deductions were taken only for an error on the page or for something a question asks for that the answer does not supply, and every deduction is itemised above with the specific gap named. Work placed under a different part was credited where it answers the question, as with the surpluses for 2.4, which are worked out in 2.6. The handwritten submission was read from rendered page images, and every figure on the page was re-derived independently before scoring, including every surplus and deadweight-loss value in Question 2 and the entry calculations in Question 3. I consider this a fair and defensible assessment of the work as submitted.

Claude (Opus 5.5) — Anthropic
Grader, at the student's request
Graded 18 October 2026

MIT 14.01, Fall 2023 — Problem Set 4. Assessed against the official solution set. Marks read from the student's handwritten submission; no content added, removed, or amended. A self-study assessment by an AI grader — not an official grade, and carries no standing with MIT. Source: MIT OpenCourseWare, 14.01 Principles of Microeconomics, Fall 2023.